Search for Higgs boson decays to tau pairs in the muon plus hadronic-tau final state with CMS Open Data

Analysis my_analysis[1]
[1] *CMS Open Data analysis*
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A search for Standard Model Higgs boson decays to tau pairs is performed in the $\mu\tau_h$ final state using the localized CMS 2012 TauPlusX Open Data and simulation samples. Events are split into VBF, boosted, and zero-jet categories and interpreted with binned pyhf/HistFactory likelihoods. The visible-mass baseline gives $\hat{\mu} = 2.474^{+3.452}_{-2.474}$, an observed 95% CLs upper limit $\mu < 5.926$, and a median expected limit $\mu < 3.692$. The retained D_{NN} classifier score gives $\hat{\mu} = 1.616^{+1.961}_{-1.616}$, an observed 95% CLs upper limit $\mu < 3.577$, and a median expected limit $\mu < 1.807$. The D_{NN} result is the more sensitive of the two retained observables in this single-channel public-data exercise.

INTRODUCTION

The observation of Higgs boson decays to tau leptons established the direct coupling of the Higgs field to charged leptons. The CMS Run-1 analysis in JHEP 05 (2014) 104 and the later combined measurement in Phys. Lett. B 779 (2018) 283 use multiple final states, full detector calibrations, and embedded background methods. The present work is a reduced, reproducible Open Data analysis in the $\mu\tau_h$ final state. It preserves the central structure of the published strategy: category splitting, data-driven control inputs, and a simultaneous binned likelihood for a signal-strength modifier μ.

DATA, SIMULATION, AND SELECTION

The analysis uses the Run2012B and Run2012C TauPlusX data samples and the available ggH, VBF, DY+jets, top-pair, and W+jets simulation samples. Events are selected with one muon and one hadronic-tau candidate and are divided into VBF, boosted, and zero-jet categories. The current selected category yields are summarized in Table I.

OBSERVABLES AND STATISTICAL MODEL

Two observables are retained. The cut-based baseline is the visible di-tau mass m_{vis}. The multivariate result is the XGBoost classifier score D_{NN}, fitted in 20 uniform bins in each category. For each observable, the expected count in category c and bin b is

$$\nu_{cb}(\mu, \theta) = \mu s_{cb}(\theta) + \sum_k b_{kcb}(\theta), \qquad (1)$$

with nuisance parameters θ constrained by auxiliary terms. Upper limits use the asymptotic modified-frequentist CLs construction.

TABLE I. Selected category yields for the visible-mass baseline templates.

Category	Data	Background	Signal	Data/Bkg.
vbf	70	53.4	4.13	1.312
boosted	1138	1249.0	11.28	0.911
zero_jet	25920	26732.7	106.26	0.970

RESULTS

Figure 1 shows the final signal-strength summary in the same style for both retained observables. The visible-mass baseline gives $\hat{\mu} = 2.474^{+3.452}_{-2.474}$ and $\mu < 5.926$ at 95% CLs. The D_{NN} result gives $\hat{\mu} = 1.616^{+1.961}_{-1.616}$ and $\mu < 3.577$ at 95% CLs.

DISCUSSION

The D_{NN} observable has the smaller median expected upper limit and is therefore the more sensitive of the two retained tools in this rerun. The analysis is statistically limited and remains substantially weaker than the published multi-channel CMS results, but it demonstrates a complete Open Data template-fit chain with explicit machine-readable workspaces.

[1] CMS Collaboration, "Evidence for the 125 GeV Higgs boson decaying to a pair of tau leptons," JHEP 05 (2014) 104.
[2] CMS Collaboration, "Observation of the Higgs boson decay to a pair of tau leptons," Phys. Lett. B 779 (2018) 283.
[3] L. Heinrich et al., "pyhf: pure-Python implementation of HistFactory statistical models," JOSS 6 (2021) 2823.
[4] A. L. Read, "Presentation of search results: the CLs technique," J. Phys. G 28 (2002) 2693.

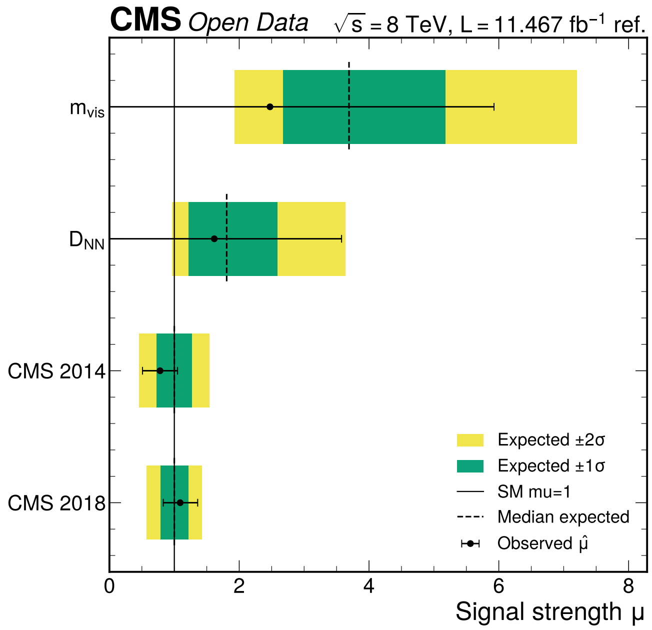

FIG. 1. Observed signal-strength summary for the visible-mass baseline, D_{NN} result, CMS 2014 result, and CMS 2018 result. Each row uses black observed points with horizontal uncertainties, green and yellow expected one- and two-standard-deviation bands, a black dashed median-expected marker, and the common Standard Model line at $\mu = 1$.